

Beam-Domain Channel Estimation for mmWave MIMO using Sub-6 GHz Out-of-Band Information

F. Pasic, M. Mussbah, S. Schwarz and M. Rupp

ISWCS, August 24-26, 2026 - Gold Coast, Australia



Technische
Universität Wien

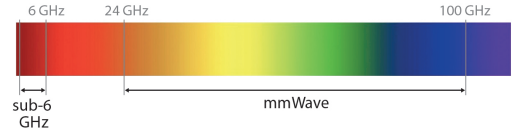


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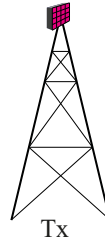


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- sub-6 GHz band cannot meet these requirements
- mmWave bands allow for significantly higher data rates

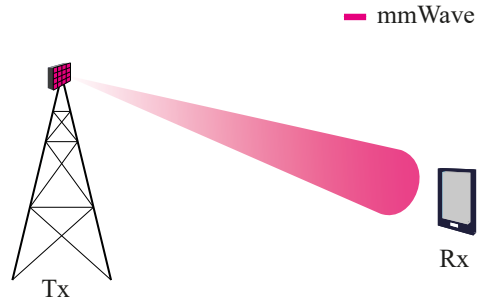


- mmWave systems utilize antenna arrays

— mmWave



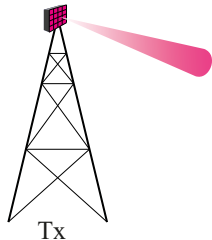
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 - apply MIMO techniques



MIMO ... multiple-input multiple-output

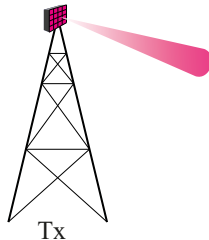
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 - **problem**: challenging link establishment



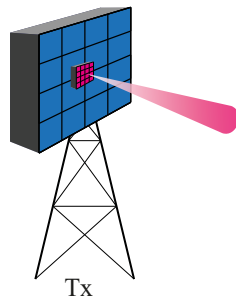
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Rx

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 - **problem:** challenging link establishment
- mmWave deployment in conjunction with sub-6 GHz systems
 - sub-6 GHz → favorable propagation conditions

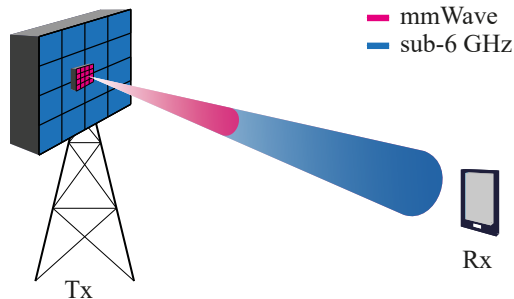


— mmWave
— sub-6 GHz

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 - **problem**: challenging link establishment
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 - sub-6 GHz → favorable propagation conditions
 - ▷ **use out-of-band channel information to improve mmWave link establishment**



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- **novel beam-domain channel estimation method for mmWave MIMO communication systems**
 - estimate mmWave channel with the aid of beam-domain sub-6 GHz out-of-band channel estimate

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 - estimate mmWave channel with the aid of beam-domain sub-6 GHz out-of-band channel estimate
- **demonstrate performance through numerical simulations**
 - achievable spectral efficiency (SE)
 - varying channel conditions in terms of Rician K -factor

System Model

Simulation Results

Summary and Outlook

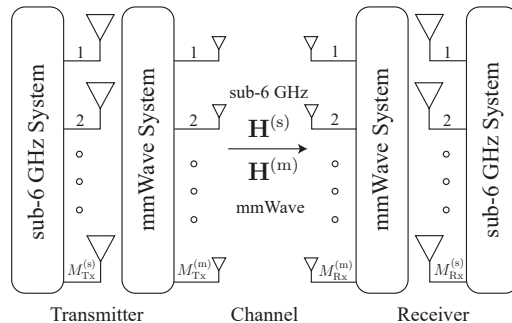
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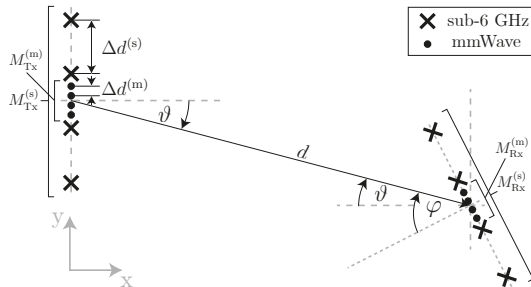
Summary and Outlook

- multi-band MIMO OFDM system

OFDM ... orthogonal freq.-division multiplexing



- multi-band MIMO OFDM system
 - co-located and aligned ULAs

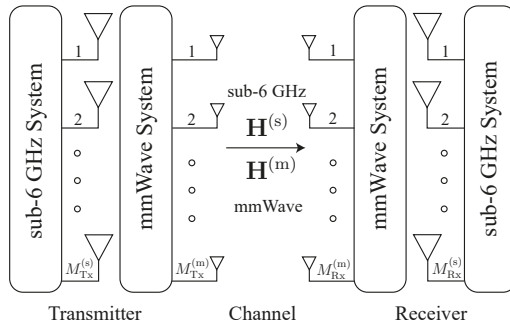


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ULA ... uniform linear array

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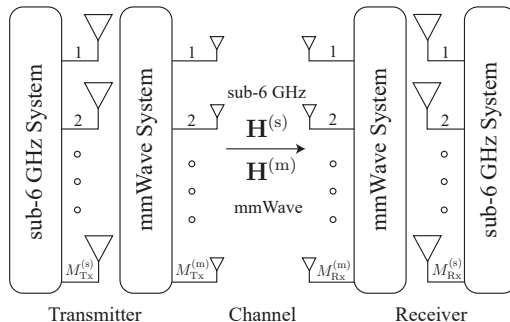


- multi-band MIMO OFDM system
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 - TDD system

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TDD ... time-division duplex



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 - Ricean channel model

$$\mathbf{H}^{(s)} = \underbrace{\sqrt{\eta^{(s)}}}_{\text{pathloss}} \left(\underbrace{\sqrt{\frac{\kappa^{(s)}}{1 + \kappa^{(s)}}}}_{\text{free-space}} \mathbf{H}_{\text{fs}}^{(s)} + \underbrace{\sqrt{\frac{1}{1 + \kappa^{(s)}}}}_{\text{stochastic part}} \mathbf{H}_{\text{sp}}^{(s)} \right)$$
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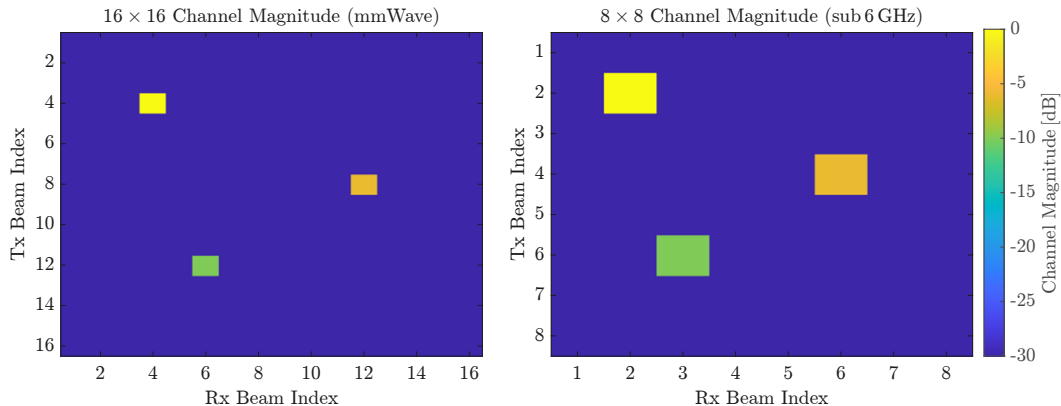
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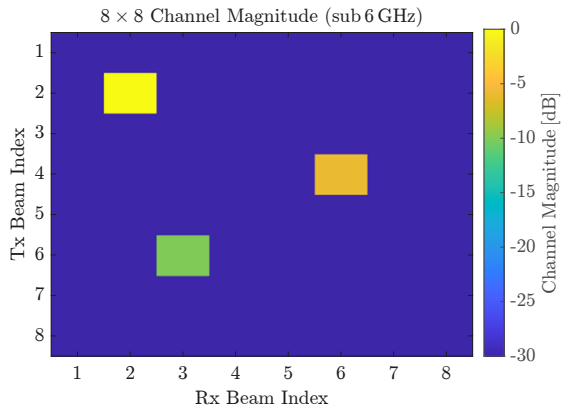
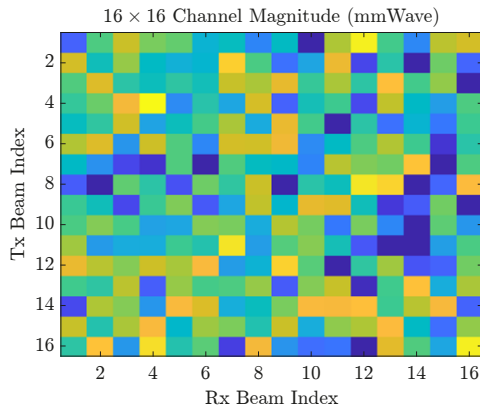
- training phase
 - both bands used: sub-6 GHz and mmWave

- data transmission phase
 - used mmWave band only
 - transmit data using precoder and combiner

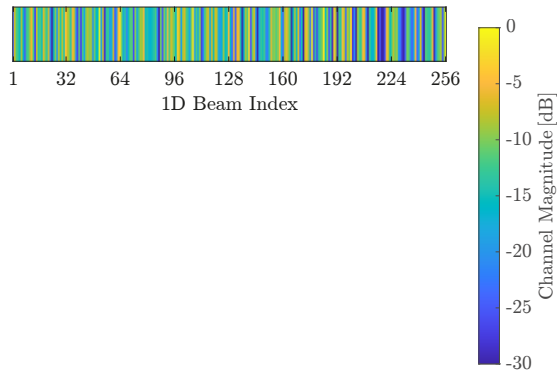
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 - resulting channel estimate $\implies$ calculate precoder and combiner
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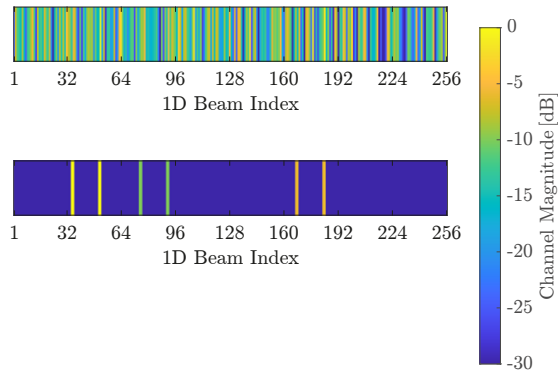




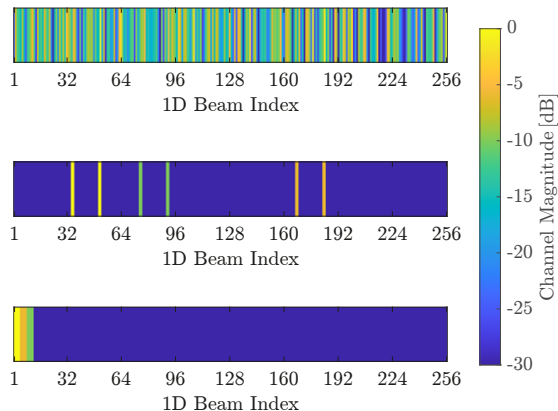
- mmWave beam-domain CSI
 - low SNR regime
 - challenging beam direction estimation



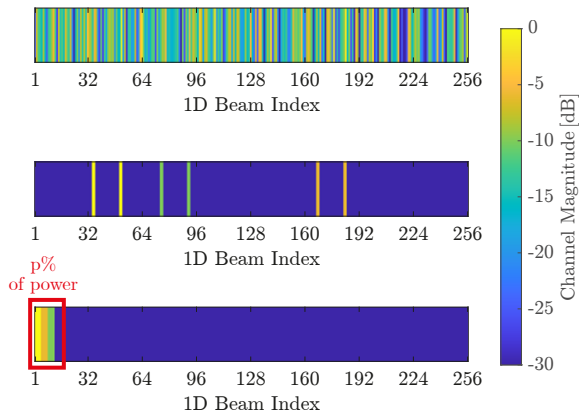
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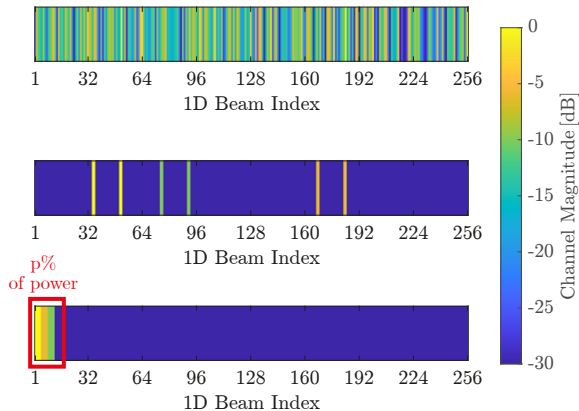
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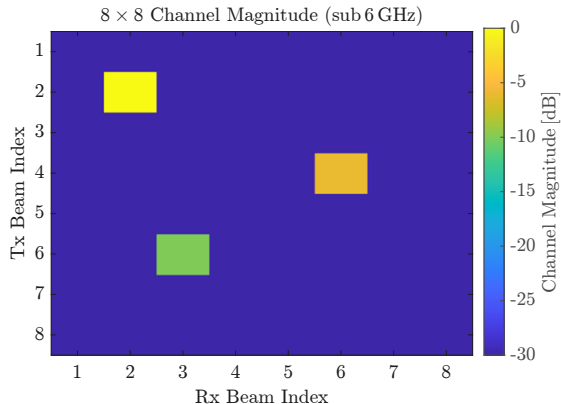
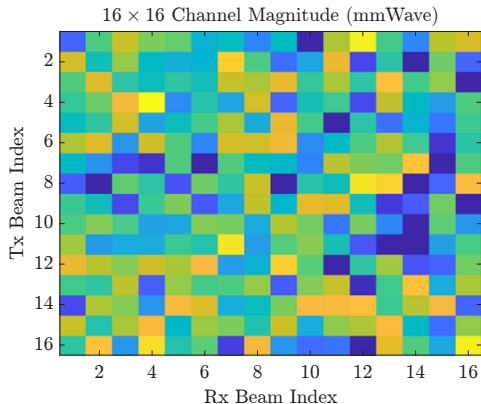


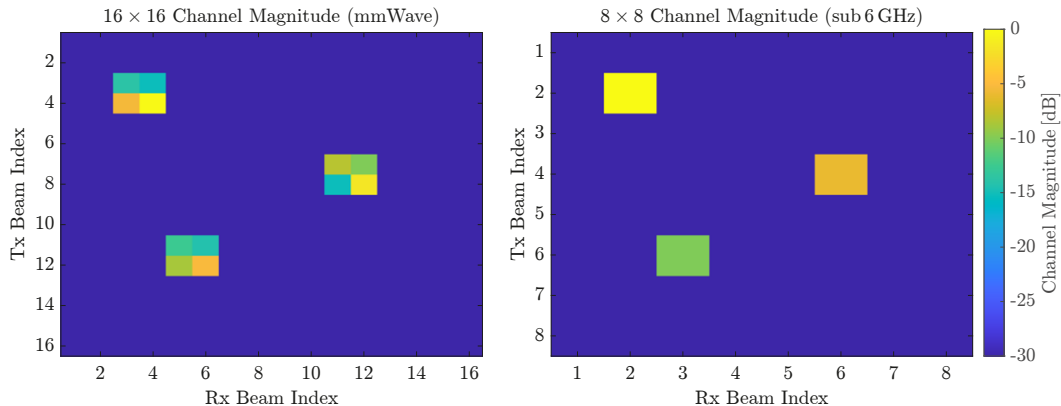
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 1. sort sub-6 GHz beams based on power
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e.g., $p = 90\%$, $p = 95\%$, ...
 3. filter the mmWave beam-domain CSI
based on the selected sub-6 GHz beams







System Model

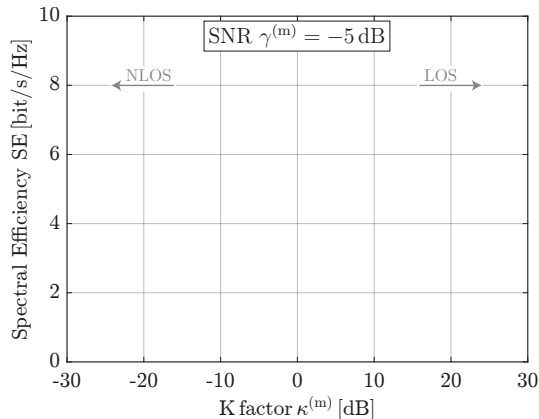
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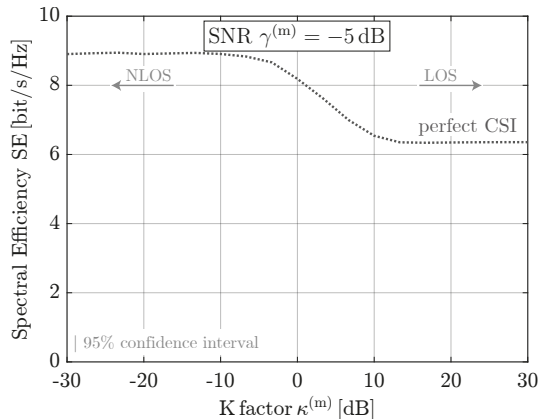
- investigate achievable spectral efficiency

Parameter	Value	
Frequency Band	sub-6 GHz	mmWave
Carrier Frequency f_c	2.55 GHz	25.5 GHz
Wavelength λ	11.76 cm	1.176 cm
Bandwidth B	20.16 MHz	403.2 MHz
Subcarrier Spacing Δf	60 kHz	120 kHz
MIMO Configuration	8×8	16×16
Transmit Power P_T	30 dBm	30 dBm
Noise Figure F	3 dB	6 dB
Number of Realizations L_r	1000	1000

- investigate achievable spectral efficiency

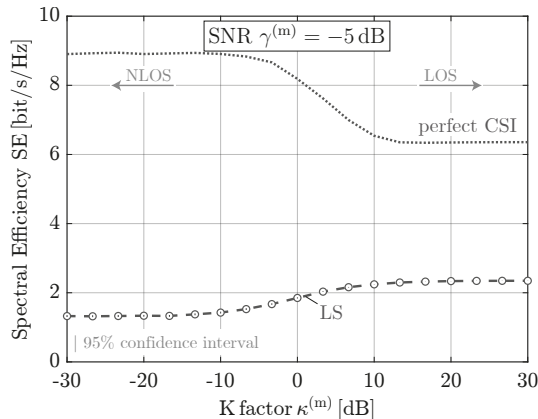


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- perfect CSI



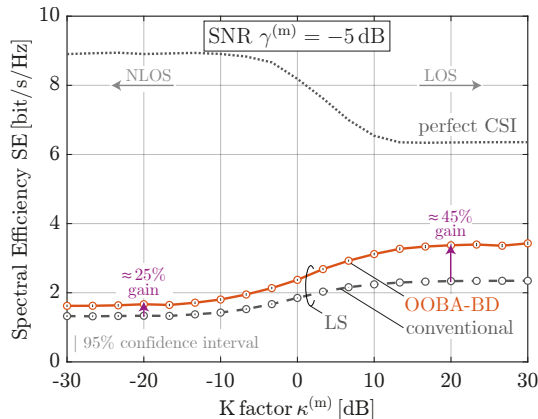
CSI ... channel state information

- investigate achievable spectral efficiency
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- least-square (LS) estimation
 - conventional method



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 - ◆ proposed method: OOBA-BD
 - NLOS scenario → **25% gain**
 - LOS scenario → **45% gain**

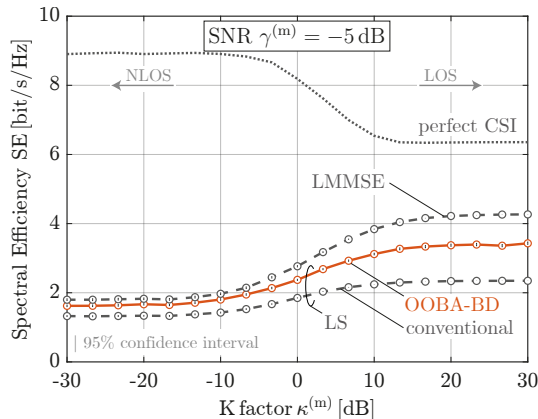


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CSI ... channel state information

LMMSE ... linear minimum mean squared error



- investigate achievable spectral efficiency

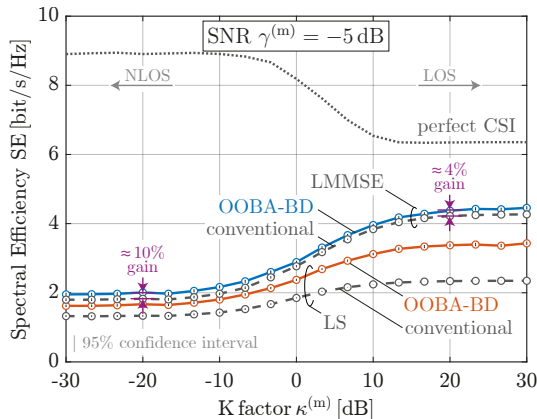
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- LMMSE estimation
 - conventional method
 - proposed method: OOBA-BD
 - NLOS scenario → **10%** gain
 - LOS scenario → **4%** gain

CSI ... channel state information

LMMSE ... linear minimum mean squared error



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- **exploiting sub-6 GHz out-of-band information within beam-domain estimation improves performance**
 - ◆ performance gains are possible irrespective of the considered K -factor
 - ◆ higher performance gains achieved in LOS scenarios compared to NLOS scenarios

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- △ **outlook:**
 - apply machine learning (ML) algorithms to further enhance performance gains
 - validate the effectiveness of the proposed method through channel measurements

Thank you for your attention!
Questions?

Faruk Pasic
faruk.pasic@tuwien.ac.at